Safe Step Prevention: Stopping Kumbh Mela Station Overcrowding

PROBLEM STATEMENT

How can we detect dangerous overcrowding at a railway platform or footbridge in real time, and **automatically prevent further entry before it becomes fatal**, rather than relying only on staff noticing the danger by eye?

The problem on the ground

- During mass gatherings such as the Kumbh Mela, stations near the event become dangerously crowded as pilgrims arrive and leave in huge numbers over a short window.
- Footbridges, platforms and entry gates were never designed to absorb a crowd of this scale.
- Both major station incidents (2013, 2025) happened despite heavy police presence. The failure was not manpower. It was **missing real-time data on crowd density and structural load** reaching the people who could act in time.

Who it affects

- Kumbh Mela pilgrims, especially **women, children and the elderly**, who are least able to move once a crowd surge starts
- Railway and station staff who must make split-second calls
- Smaller stations and footbridges around Kumbh sites with no camera/drone command centre

Why Kumbh Mela

The Kumbh Mela brings pilgrims in extremely large numbers over a short period, and the railway stations and footbridges leading to it take the load. Two of the deadliest recent incidents, in 2013 and 2025, happened at Kumbh-linked railway stations.

Why It Matters, and the Gap in Today's Systems

Reported incidents (estimates)

Incident	Deaths	Cause
1954, Allahabad Kumbh Mela	~50	General crowd stampede
2013, Allahabad Station foot overbridge	~36	Bridge railing collapsed from overcrowding
Jan 2025, Maha Kumbh (Sangam)	~30	Crowd crush at bathing area
Feb 2025, New Delhi Railway Station	~18	Stampede boarding trains to Kumbh

Two of these four incidents happened on exactly the structures Safe Step Prevention targets: a railway footbridge and a railway station.

What authorities already do

- Holding shelters and phased boarding; platform tickets suspended at peak
- Separate entry/exit routes and wider footbridges with ramps
- AI-CCTV and drones feeding a central control room that alerts staff
- Gates closed or crowds diverted **manually** once alerted

The gaps we found

- **Limited reach.** Hub-station infrastructure (Prayagraj, Ayodhya, Varanasi) needs a very large budget and is used mainly for the Maha Kumbh. Thousands of smaller gatherings get none of it.
- **No structural sensing.** Existing systems estimate crowd density from cameras. None we found measures physical strain on a bridge, the cause of the 2013 collapse.
- **Human delay.** A person must decide and act, so there is a gap between an AI alert and a gate actually closing.

Safe Step Prevention: Automatic Gate Control for One Checkpoint

A **low-cost, automatic** system for a single platform or footbridge. It is not a replacement for large government systems. It is an affordable version of the same safety principle, with two improvements existing systems lack.

Live occupancy count

Tracks how many people are on the platform or bridge right now, not just total footfall.

Direct structural load

A load cell measures real strain on the deck, an independent check that does not rely on crowd-size estimates.

Instant physical barrier

The gate locks the moment a threshold is crossed. It works even where loud, panicked crowds cannot hear announcements.

Staff stay in control

Dashboard shows live data. Staff keep full manual override at all times. Safe Step Prevention assists judgment, never replaces it.

Technology Stack, Prototype Budget & Safety by Design

Prototype components (approx. Rs. 1,900 total)

Component	Purpose	Cost
Arduino Uno	Platform/bridge sensor node	Rs. 500
Arduino Nano	Central hub	Rs. 350
HC-SR04 ultrasonic x2	Entry + exit people counting	Rs. 140
Load cell (20 kg) + HX711	Bridge structural strain	Rs. 250
Micro servo motor	Automatic gate arm	Rs. 150
nRF24L01+ x2	Wireless link between nodes	Rs. 200
Buzzer + LED	Local alert	Rs. 30
Breadboards, wires, solder	Assembly and testing	Rs. 250-300

Software: Arduino IDE (C/C++), open-source sensor libraries, public GitHub repository with README, architecture notes and sample data.

The servo is a basic stand-in to demonstrate the concept. Real deployment needs industrial-grade actuators.

Safety by design: known risks and fixes

- **Gate must never trap people.** Fail-safe: it opens automatically on power loss or an emergency trigger.
- **Sudden closure can push the crowd.** Slow the entry rate gradually, add a holding area before the gate, and warn well before the limit.
- **Sensors can undercount in dense crowds.** Treat data as a strong signal, pair with periodic manual headcounts.
- **Prototype is proof of concept.** Station-scale rollout needs engineering review, safety testing and phased deployment.

Expected Impact, Measurable KPIs and Scalability

Expected outcomes

- **Faster than manual systems:** the gate acts the instant a threshold is crossed, with no wait for a human decision.
- **Catches what cameras miss:** the load cell targets the structural overload behind the 2013 bridge collapse.
- **Works in chaos:** a physical barrier does not depend on people hearing or obeying a warning.
- **Protects the vulnerable:** women, children and the elderly are most at risk in crowd surges.

Measurable KPIs (to validate in pilot)

- Time from threshold crossed to gate closed (design target: under 1 second)
- Peak occupancy kept at or below the set safe limit
- Over-threshold events detected and entries blocked
- Sensor count accuracy vs manual headcount (%)
- Structural load readings staying under the deck's rated limit
- Cost per checkpoint (prototype about Rs. 1,900)

Scalability & sustainability

- **Modular:** one node per checkpoint. Add nodes for more platforms, stairs or gates.
- **Beyond Kumbh:** any high-footfall footbridge, platform, staircase, temple or stadium entrance.
- **Any region or language:** a barrier, buzzer and LED need no language. Dashboard text can be localised.
- **Low cost, open source:** off-the-shelf parts and public code that other colleges and local bodies can copy.

Participant Profile and Responsibilities

INDIVIDUAL PARTICIPANT

Rakshitha D

Department of Electrical and Electronics Engineering (EEE)

State: Tamil Nadu

Theme: Disaster Mitigation & Resilience

I am taking Safe Step Prevention forward as a solo participant and own every part of the project, from hardware to pitch.

Roles I cover end to end

Hardware	Occupancy sensors and load cell (HX711) wiring, calibration and testing
Firmware	Automatic gate servo control and wireless link between sensor node and hub
Hub logic	Threshold rules for occupancy and load, gate control, fail-safe behaviour
Dashboard	Staff dashboard for live readings, alerts and manual override
Pitch & demo	Presentation deck, 3-5 minute demo video and the public GitHub repository

Support I would value

- Mentoring from IIT Bombay faculty and experts on structural-load calibration and safety testing
- Guidance from station or festival authorities for a small supervised pilot

Implementation Roadmap and Resource Requirements

PHASE 1
Submission & screening

Now to Oct 25, 2026

- Finalise deck, README and architecture notes
- Publish public GitHub repository
- Record demo video (preferred)

PHASE 2
State demo / pitch

Oct 26 to Nov 30, 2026

- Build breadboard prototype: counting, load cell, servo gate
- Demonstrate live trigger and gate lock
- Pitch to the state jury

PHASE 3
If National Finalist

Dec 16, 2026 to Jan 9, 2027

- Use Rs. 6,000 grant to improve the prototype
- Calibrate thresholds, add fail-safe gate-open
- Mentor sessions on safety validation

PHASE 4
Showcase & pilot

Jan 10-12, 2027 and beyond

- Exhibit at VBYLD 2027, 3-minute pitch
- Propose a small supervised pilot
- Engineering review, then phased rollout

Resource requirements

- **Hardware:** about Rs. 1,900 for the prototype (see slide 4)
- **Software:** Arduino IDE, open-source libraries, GitHub (all free)
- **Expertise:** mentor review of load-cell calibration and gate fail-safe design
- **Permissions:** station or festival authority approval before any real-world pilot